Sahil Ali Malik | Mechanical Design & Product Development Engineer

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PROFILE

Results-driven Mechanical Design and Product Development Engineer with 6 years of experience driving end-to-end product development in fast-paced startup environments across robotics, AgriTech, automotive, consumer electronics, and medical equipment. Proven ability to translate concepts into production-ready hardware, solve complex engineering challenges, and optimize product design for performance, reliability, and manufacturability. Strong technical expertise in detailed engineering, SOLIDWORKS, DFMA, GD&T, fits and clearances, tolerance stack-up analysis, mechanism design, sheet metal design, and rapid prototyping. Proven ability to independently own mechanical design from concept through manufacturing, collaborate with cross-functional teams and manufacturing partners, and drive engineering improvements that enhance product quality, manufacturability, reliability, and production efficiency.

PROFESSIONAL EXPERIENCE

Mechanical Design Engineer | CRTS Global, Al Khobar, Saudi Arabia [Apr 2026-Present]

Designing and developing robotic coating systems, owning mechanical product development from concept and system architecture through detailed engineering, prototyping, testing, and manufacturing.

- Drive mechanical redesign and optimization of existing robotic platforms, resolving functional, assembly, and manufacturability challenges to improve machine performance and reliability.
- Develop new robotic platforms and mechanical architectures, including structural frameworks, mechanisms, enclosures, mounting systems, and critical machine assemblies.
- Leverage in-house machining, welding, and 3D printing capabilities to accelerate prototyping, design iterations, fit checks, and mechanical validation.
- Apply GD&T, fits and clearances, tolerance stack-up analysis, and DFM/DFA principles to deliver manufacturing-ready designs and ensure effective integration with manufacturing and assembly teams.

Mechanical Design Engineer | Niqo Robotics, Bangalore, India [Sep 2024-Mar 2026]

- Led mechanical design and new product development for AI-driven agricultural automation systems, translating system-level requirements into robust, production-ready mechanical solutions.
- Designed and developed critical subsystems for the flagship **Robothinner** platform, including structural assemblies, hydraulic component mounting systems, protective enclosures, and mechanical interfaces for integrated automation.
- Engineered **IP-rated enclosures and customized packaging solutions** for electronic components, cameras, and sensors, balancing environmental protection, functional performance, assembly accessibility, and manufacturability.
- Applied **GD&T, tolerance stack-up analysis, fits and clearances, and DFM/DFA principles** to complex assemblies, resolving fitment challenges and improving manufacturing accuracy, assembly consistency, and product quality.
- Collaborated with electrical, software, manufacturing, and assembly teams to integrate mechanical hardware with electronics, sensing systems, and software-driven automation requirements.
- Participated in **Design FMEA** to identify potential failure modes, assess design risks, and implement mitigation measures to improve product reliability and operational performance.
- Managed engineering changes for products supporting high-volume manufacturing, ensuring design updates were properly documented, validated, and implemented in line with applicable regulatory and production requirements.
- Standardized engineering documentation and drawing practices in alignment with **ASME Y14.5**, establishing consistent CAD release procedures, document storage structures, and access protocols to improve traceability and reduce communication errors.
- Worked closely with internal teams and external manufacturing partners to resolve design-for-manufacturing challenges, support prototype builds, conduct fit checks, and improve first-time manufacturing outcomes.

Design Engineer – Sustaining | Inito Health, Bangalore, India [Aug 2022-July 2024]

- Drove product development and improvement initiatives for home diagnostic test kits and monitoring devices, resolving hardware issues, customer complaints, and design-related quality concerns.
- Designed and prototyped an automated incoming quality-control system for Aztec Code verification, developing conveyor, strip-alignment, inspection, and rejection mechanisms to improve inspection throughput and reliability.
- Conducted **Design Failure Mode and Effects Analysis (DFMEA)** for New Product Introduction (NPI) and supported Root Cause and Corrective Action (RCCA) activities to identify failure modes, resolve product and process issues, and improve product quality.
- Managed the **Engineering Change Notice (ECN)** process, coordinating cross-functional reviews and ensuring accurate implementation of design changes in accordance with applicable regulatory requirements and internal procedures.
- Applied **DFM/DFA, GD&T, tolerance stack-up analysis, and rapid prototyping** to improve manufacturability, assembly efficiency, and product performance while accelerating design iteration and validation.

- Provided technical support to contract manufacturers and assembly partners, conducting manufacturing audits, verifying compliance with engineering drawings and SOPs, and implementing corrective actions to maintain quality standards.
- Designed production jigs and fixtures to improve assembly repeatability, dimensional accuracy, and process efficiency while addressing manufacturing challenges in collaboration with production teams.
- Collaborated with Production, Quality Engineering, Operations, and Packaging teams to prioritize engineering activities, support business continuity, and develop functional, protective, and regulation-compliant packaging solutions for diagnostic products.

Mechanical Design Engineer | Hydax Hydraulics Pvt. Ltd, Bangalore, India [Apr 2021-Aug 2022]

- Designed and developed a prototype **automated mechanical deburring machine** for gear finishing, contributing to concept development, detailed engineering, manufacturing, assembly, and functional validation.
- Developed mechanical design solutions and performed design analysis using **SOLIDWORKS**, translating customer requirements into practical, manufacturable engineering solutions.
- Created and converted 2D and 3D CAD models and generated detailed manufacturing drawings for customers, suppliers, and production teams, applying **DFMA principles** to support efficient manufacturing and assembly.
- Prepared bills of materials (BOMs), verified finished-part dimensions against engineering drawings, and incorporated design changes based on customer feedback and manufacturing requirements.
- Coordinated with external suppliers, solution providers, and manufacturing vendors to source components, manage procurement requirements, and support timely project execution.
- Monitored project progress, communicated engineering updates to customers and internal teams, prepared project proposals, and delivered technical presentations during customer interactions.
- Resolved engineering challenges in hydraulic component manufacturing through practical design improvements while coordinating design changes, manufacturing requirements, and supplier deliverables.

Design Intern | Technologics Global Pvt. Ltd, Bangalore, India [Jan 2021- Mar 2021]
Developed proficiency in CAD software, including SOLIDWORKS, CATIA V5, and Mastercam, while contributing to the design and development of special-purpose machines (SPMs) and applying product development fundamentals to practical engineering projects.

Intern | DeltaT MEP Technical Academy, Bangalore, India [Jan 2020- Feb 2020]
Worked on HVAC system architectures and structural analysis, designing and optimizing components for performance, reliability, and manufacturability while utilizing industry-standard CAD and simulation software for component modeling and analysis.

EDUCATION

Visveswaraiah Technological University, Karnataka, India *2016 - 2020*
Bachelor of Engineering (B.E.) in Mechanical Engineering **CGPA: 8/10**

KEY PROJECTS

Roboweeder Solo – Lightweight Single-Bay Robotic Weeder *[Associated With Niqo Robotics]*

- Owned end-to-end mechanical design of a lightweight, single-bay robotic weeder adapted from the existing three-bay platform for the Indian and Australian markets.
- Designed the machine architecture, chassis, welded structures, protective skirting, adjustable camera mounting system, electronics hood, hitch points, and manifold actuation mechanism.
- Defined fits, clearances, and tolerance stack-ups; developed manufacturing-ready drawings with GD&T for complex welded and machined components.
- Collaborated with cross-functional and manufacturing teams to develop prototypes, resolve assembly challenges, and refine designs for manufacturability.

Impact: Developed the mechanical architecture and initial prototype of a lightweight single-bay Roboweeder platform. The project involved system-level mechanical design, structural engineering, GD&T, tolerance definition, prototyping, manufacturing coordination, and cross-functional problem solving from concept through manufacturing.

Niqo Sense Camera Design – IP69K Rated Camera *[Associated With Niqo Robotics]*

- Redesigned the camera enclosure to accommodate updated PCB and electronic components while addressing waterproofing, packaging, and assembly requirements.
- Conducted waterproofing experiments to identify leakage paths and developed a dovetail-groove housing concept to evaluate sealing performance.
- Engineered the enclosure with integrated sealing features, optimizing fits, clearances, and assembly through iterative prototyping and fit-check feedback.

- Coordinated with cross-functional teams and external machining vendors to accelerate prototyping, resolve design issues, and finalize manufacturing-ready components.

Impact: Achieved IP69K certification through laboratory testing, upgrading a previously non-IP-rated camera while accommodating revised electronics and improving assembly within a tight development timeline.

RTK Base Station Design – RF & IP67 Optimization *[Associated With Niqo Robotics]*

- Owned mechanical development from concept through mass production, redesigning the RTK Base Station enclosure to address signal-range issues while achieving IP67 protection.
- Optimized enclosure architecture and surrounding material selection to improve RF transparency while maintaining environmental sealing.
- Performed tolerance stack-up analysis on critical assembly interfaces, achieving LED-to-enclosure positioning within ±0.5 mm in the initial design iteration.
- Collaborated with assembly teams to resolve fitment issues, refine design iterations, and support manufacturing readiness.

Impact: Achieved IP67 protection while addressing RF performance challenges, delivering a production-ready design within the timeline for the Roboweeder's US launch.

Actuation Mechanism for Live Manifold Control *[Associated With Niqo Robotics]*

- Performed mechanical calculations and developed a linear actuator-based mechanism to maintain consistent ground-to-nozzle distance over uneven agricultural terrain.
- Selected and integrated a LINAK linear actuator based on load, stroke, and operational requirements, incorporating laser-based depth sensing for ground-distance monitoring.
- Redesigned structural interfaces and integrated linear guides, bushings, sheet-metal mounts, and shaft bellows to support controlled movement and protect components from agricultural contamination.
- Developed detailed manufacturing and welding drawings, performed fit checks and tolerance analysis, and achieved 1 mm shaft-to-shaft alignment tolerance during machining.

Impact: Successfully implemented the live manifold actuation mechanism with the first design iteration, enabling controlled nozzle-height adjustment to accommodate uneven terrain.

Strip Qualifier – Automated QC System for Aztec Code Inspection *[Associated With Inito Health]*

- Owned mechanical design and supported manufacturing, assembly, testing, and validation of an automated quality inspection system for uniquely coded strips.
- Designed the conveyor-based strip-feeding, alignment, inspection, and automatic rejection mechanisms to streamline material handling and inspection.
- Developed adjustable mounting structures with three degrees of freedom for precise positioning of cameras, dome lighting, and sensors.
- Selected and sourced mechanical components, coordinated with external manufacturing vendors, and collaborated with the assembly team to resolve integration and troubleshooting issues.

Impact: Increased QC throughput from 4,000 to 15,000 strips per day (3.75× improvement), reducing manual inspection requirements. Delivered the validated solution within three months while managing parallel engineering responsibilities, enabling deployment across three additional machines.

Jig and Fixture Design 1: Mechanical platform design for assembling the clip adapters *[Associated With Inito Health]*

- Designed and 3D-printed a precision assembly jig to standardize clip adapter assembly and minimize operator-dependent variation.
- Developed the fixture to achieve assembly accuracy of ±0.15 mm, improving repeatability and dimensional consistency.

Impact: Reduced reliance on 100% inspection by enabling a shift toward random-sampling quality control, subject to the applicable quality standards and validation requirements.

Burrgon M – Automated Gear Deburring Machine *[Associated With Hydax Hydraulics Pvt. Ltd.]*

- Owned mechanical design and contributed to end-to-end machine development, from concept and detailed engineering through manufacturing, assembly, testing, and validation.
- Designed a pneumatic tool-feed mechanism and motor-driven gearbox system for controlled workpiece rotation and automated deburring of gear cutting edges.

- Coordinated with external manufacturing vendors and the assembly team to resolve fabrication and integration challenges and validate machine functionality.
- Collaborated directly with the company's CEO throughout product development and presented the machine to prospective customers at AMTEX 2022, New Delhi.

Impact: Developed a functional automated deburring solution within four months, reducing reliance on manual burr removal and improving the gear finishing workflow.

SKILLS & ABILITIES

- **End-to-End Product Development:** Independent ownership from concept development and system architecture through detailed engineering, prototyping, testing, validation, and mass production.
- **Advanced SOLIDWORKS Expertise:** Complex 3D modeling, top-down assembly design, large assembly management, sheet metal, weldments, and detailed manufacturing drawings.
- **Complex Manufacturing Drawings:** Machining and welded fabrication drawings, GD&T (ASME Y14.5), fits and clearances, dimensional tolerancing, and manufacturing documentation.
- **Precision Engineering:** Tolerance stack-up analysis, critical interface management, fit checks, assembly validation, and dimensional accuracy.
- **Design for Manufacturing & Assembly (DFM/DFA):** Design optimization for manufacturability, assembly efficiency, reliability, serviceability, and cost-effective production.
- **Mechanical Design & System Integration:** Mechanism design, structural frameworks, IP-rated enclosures, hydraulic mounting systems, and camera, sensor, and electronic component integration.
- **Rapid Prototyping & Design Validation:** 3D printing, prototype development, iterative design improvement, functional testing, and troubleshooting.
- **Fast-Paced Project Execution:** Delivering precision-engineered solutions under tight timelines while managing design iterations, engineering changes, and manufacturing constraints.
- **Manufacturing & Supplier Collaboration:** Coordination with machine shops, fabrication vendors, contract manufacturers, and assembly teams to resolve production challenges and achieve first-time-right outcomes.
- **Engineering Quality & Change Management:** DFMEA, Root Cause Analysis (RCA/RCCA), Engineering Change Management (ECN), New Product Introduction (NPI), and PLM/CAD data management.